# Project Lost Piglet 2: Investigating Human Threat Actors via Cognitive Friction & Machine Learning

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## The Problem

Modern enterprise networks experience a daily barrage of overwhelming traffic, representing both friendly users as well as unfriendly visitors. Threats often use pre-loaded botnet programs capable of inhuman execution; presenting an ever-evolving and truly contemporary challenge for security practitioners as the world adapts to emerging technologies.

## The Objective

To design a logically-derived filter utilizing 'Cognitive Friction' psychological lures to force attackers to make decisions and revealing behavioral patterns; separating automated scanners and targeted threat activity through measurable differences in command timing and behavioral complexity.

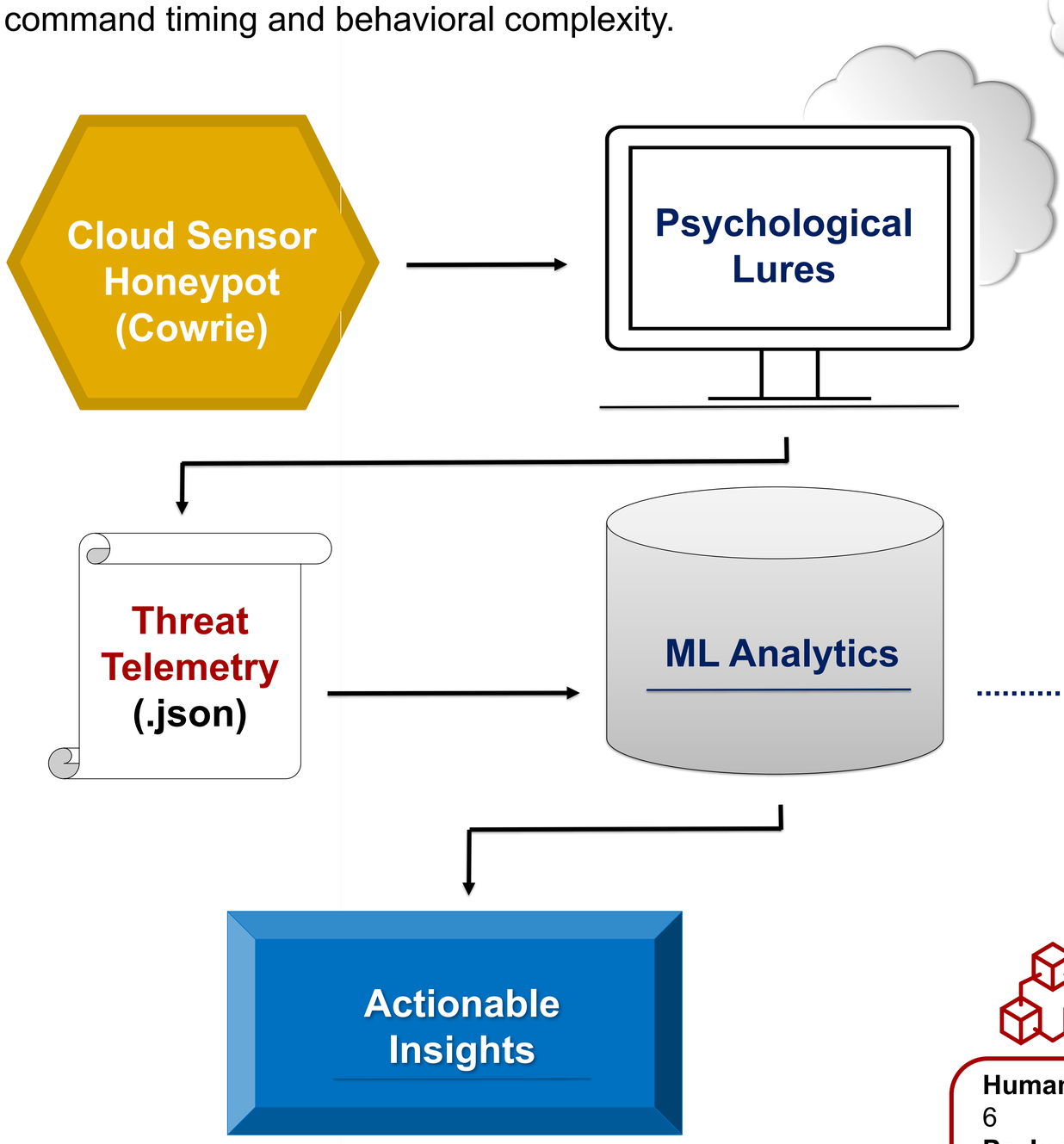


**Figure 1. Execution Strategy**

## Key Outcomes

By engineering a realistic digital deception environment, this project demonstrated that 'Cognitive Friction' can be measured and classified using ML strategies. Preliminary results suggest that behavioral modeling of hesitation is a viable mechanism for separating human threat actors from automated noise - supporting the core hypothesis.

### References

Hecker (2012) – Intelligent Honeypot Deployment
Fraunholz et al. (2018) – Survey on Deception in InfoSec.
Udhani & Withers (2019) – Human vs Bots in Honeypots
Hetzler (2024) – Cloaked Cowrie Honeypot Monitoring

### Motivating Faculty

Dr. Bartolacci | Dr. Nasereddin

### Acknowledgements

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## Cognitive Friction Methodology

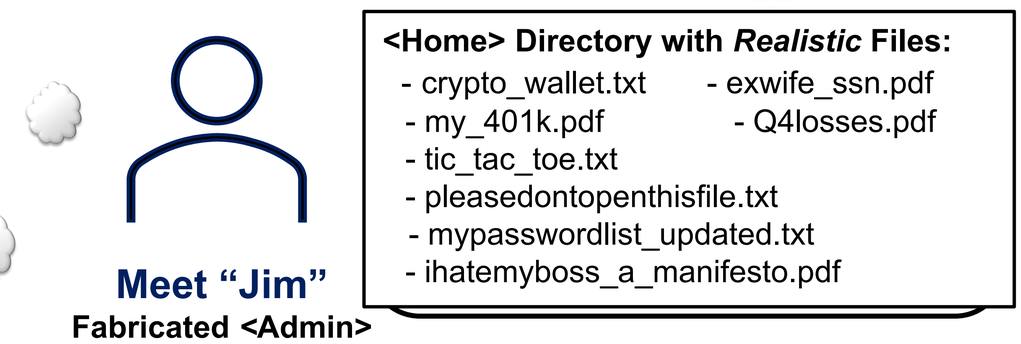


**Figure 2. High-Interaction Honeypot Configuration**

**Experiment Duration:**
648.9 hours (27 days)
**Total Events:**
174,963

**Top 3 Threat Origins:**
China (56%)
India (15.8%)
Malaysia (6.6%)

## Applied Machine Learning Methodology

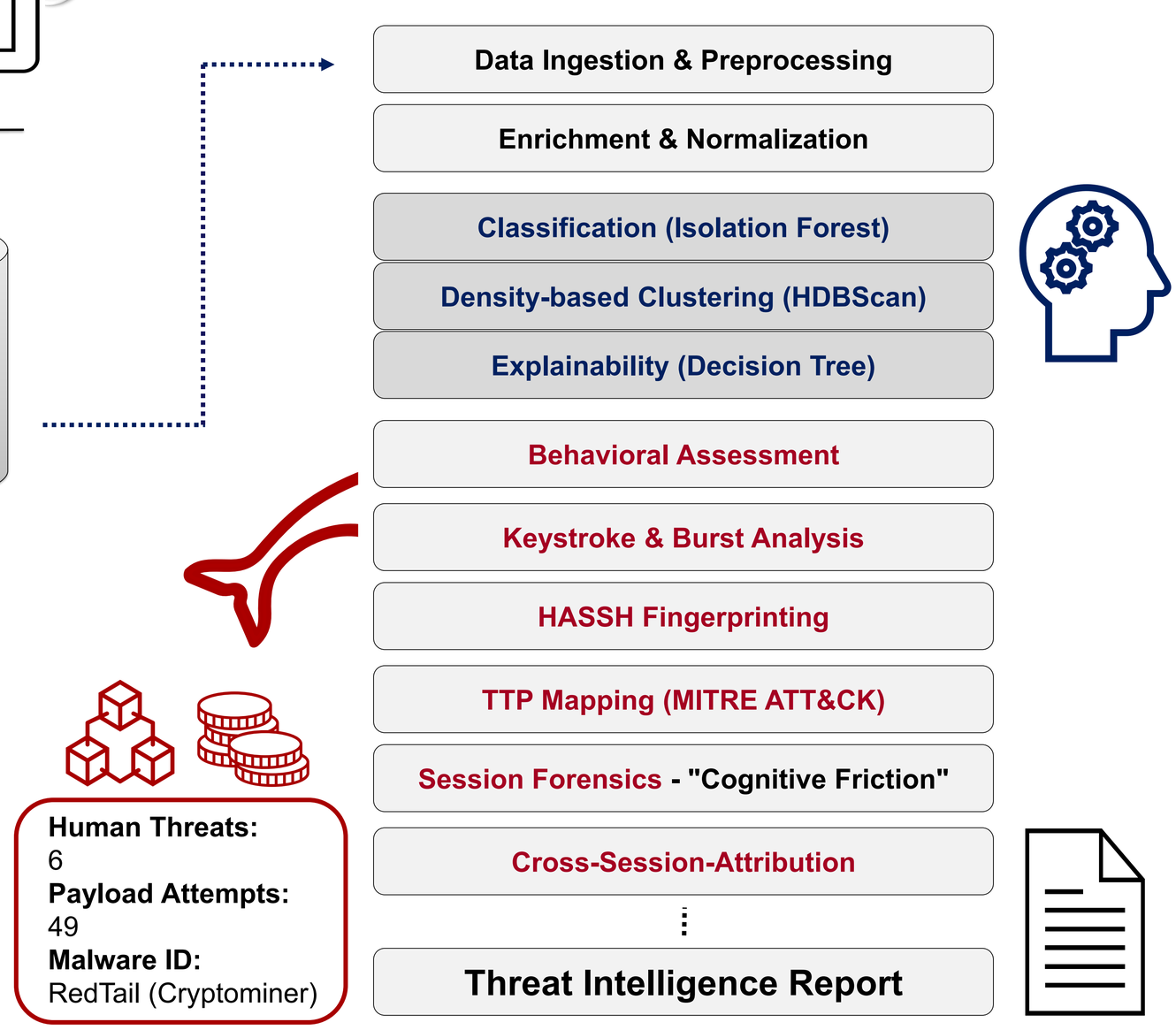


**Figure 3. ML Approach & Flow**

## Abstract

This independent cyber-intelligence initiative - "Project Lost Piglet 2 (LP2)" explores behavioral modeling and applied machine learning to isolate malicious human actors from commodity botnet traffic as well as LLM-enabled threats. Building upon cloud sensor architecture, the LP2 honeypot was engineered around the concept of 'Cognitive Friction' and deployed for collection between February 03 and March 5, 2026.

Combining observed decision-making latency with session forensics, keystroke cadence, HASSH fingerprinting, and more; the resulting assessment is a logic-derived filter capable of classifying digital threat behavior. Behavioral ML achieved 99.96%+ noise reduction across 174,963 events, isolating 6 human-interactive sessions, 6 unique payloads across 49 delivery cycles (RedTail Cryptominer), 33-day persistent botnet infrastructure across two deployments (LP1 & LP2): supporting further development of this methodology in LP3.